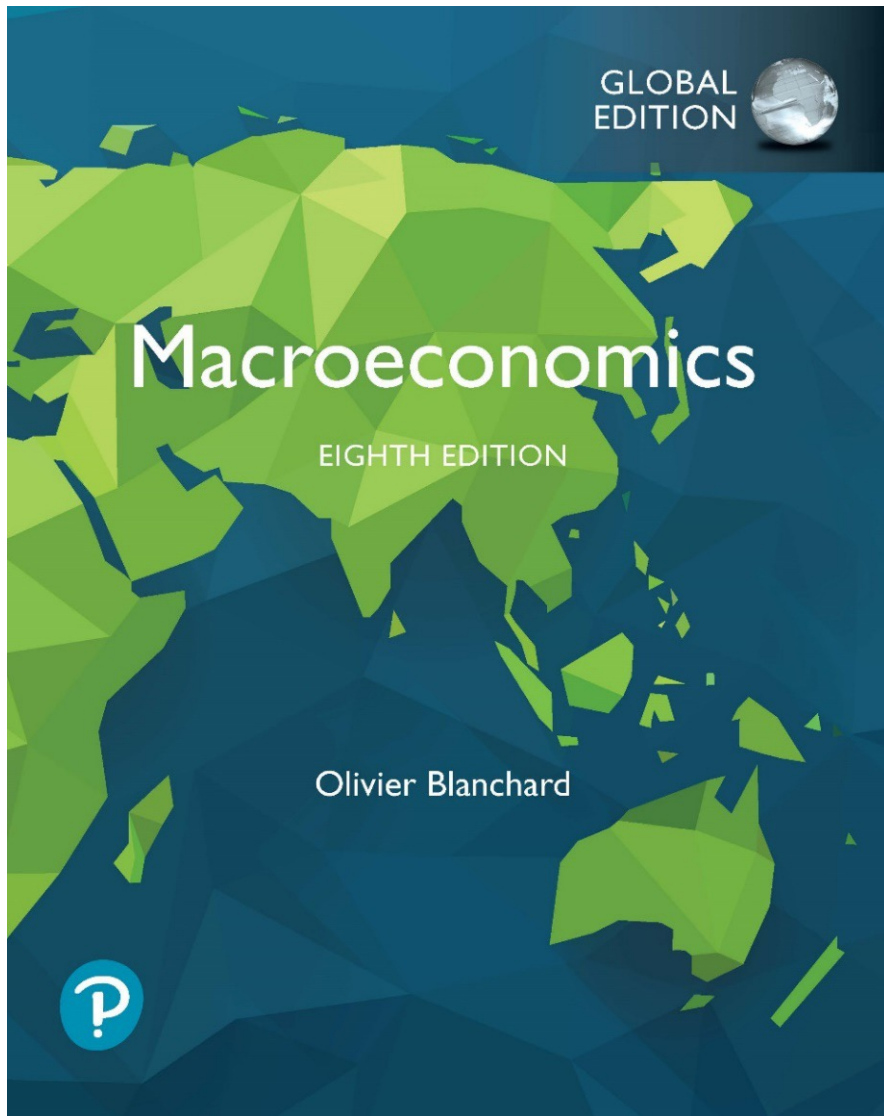


Macroeconomics

Eighth Edition, Global Edition



Chapter 14

Financial Markets and Expectations

Chapter 14 Outline

Financial Markets and Expectations

14.1 Expected Present Discounted Values

14.2 Bond Prices and Bond Yields

14.3 The Stock Market and Movements in Stock Prices

14.4 Risk, Bubbles, Fads, and Asset Prices

APPENDIX Deriving the Expected Present Discounted Value Using Real or Nominal Interest Rates

Financial Markets and Expectations

- Our focus throughout this chapter will be on the role expectations play in the determination of asset prices, from bonds, to stocks, to houses.
- We discussed the role of expectations informally at various points in the core.
- It is now time to do it more formally.

14.1 Expected Present Discounted Values (1 of 6)

- The **expected present discounted value** of a sequence of future payments is the value today of this expected sequence of payments.
- Expected present discounted values are not directly observable but must be constructed from information on the sequence of expected payments and expected interest rates.

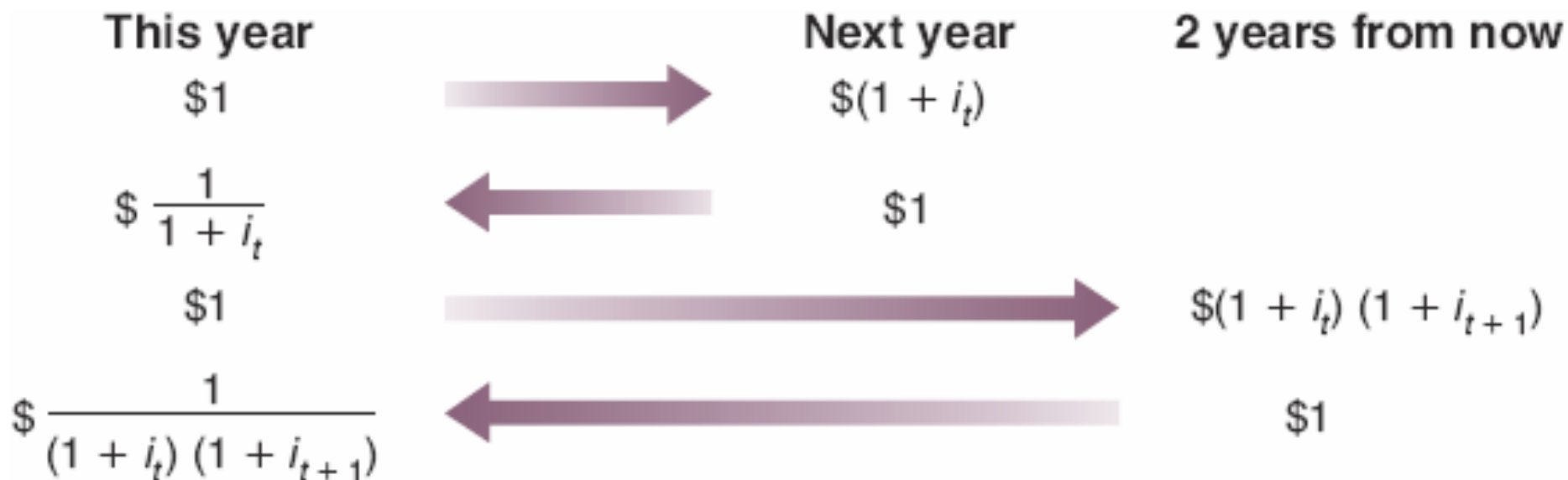
14.1 Expected Present Discounted Values (2 of 6)

- $1/(1 + i_t)$ is the **discount factor** with the **discount rate** i_t , which is used to compute the *present discounted value* of one dollar to be received next year.
- The higher the nominal interest rate, the lower the value today if a dollar received next year.

14.1 Expected Present Discounted Values (3 of 6)

- Example: The value today of a dollar received two years from now.

Figure 14.1 Computing Present Discounted Values



14.1 Expected Present Discounted Values (4 of 6)

- The **expected present discounted value** or **present value** of a payment $\$z$ and its expected payment $\$z^e$ in the future:

$$\$V_t = \$z_t + \frac{1}{(1 + i_t)} \$z_{t+1}^e + \frac{1}{(1 + i_t)(1 + i_{t+1}^e)} \$z_{t+2}^e + \dots \quad (14.1)$$

- For constant interest rates i :

$$\$V_t = \$z_t + \frac{1}{(1 + i_t)} \$z_{t+1}^e + \frac{1}{(1 + i_t)^2} \$z_{t+2}^e + \dots \quad (14.2)$$

- For constant interest rates and constant payments $\$z$:

$$\$V_t = \$z \left[1 + \frac{1}{(1 + i)} + \dots + \frac{1}{(1 + i)^{n-1}} \right]$$

14.1 Expected Present Discounted Values (5 of 6) (Assume payment start next year and forever)

$$\$V_t = \$z \left[\frac{1}{(1+i)} + \dots + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + \dots \right]$$

- For constant interest rates and **payment forever** : $\$V_t = \frac{\$z}{i}$
- If $i = 0$, then the present discounted value of a sequence of expected payments is just the sum of those expected payments.
- The present value of a sequence of real payments (discounted by real interest rates r_t):

$$V_t = z_t + \frac{1}{(1+r_t)} z_{t+1}^e + \frac{1}{(1+r_t)(1+r_{t+1}^e)} z_{t+2}^e + \dots \quad (14.3)$$

14.1 Expected Present Discounted Values (6 of 6)

- Two ways to compute the present value of payments:
 1. The present value of the sequence of payments expressed in dollars, discounted using nominal interest rates, and then divided by the price level today.
 2. The present value of the sequence of payments expressed in real terms, discounted using real interest rates.

14.2 Bond Prices and Bond Yields

(1 of 9)

- Bonds differ in two basic dimensions:
 - **Maturity**: The length of time over which the bond promises to make payments to the holder of the bond.
 - *Risk*: (1) **Default risk as the risk that the issuer of the bond will not pay back the full amount promised by the bond; or (2) price risk as the uncertainty about the price you can sell the bond for if you want to sell it in the future before maturity.**

14.2 Bond Prices and Bond Yields

(2 of 9)

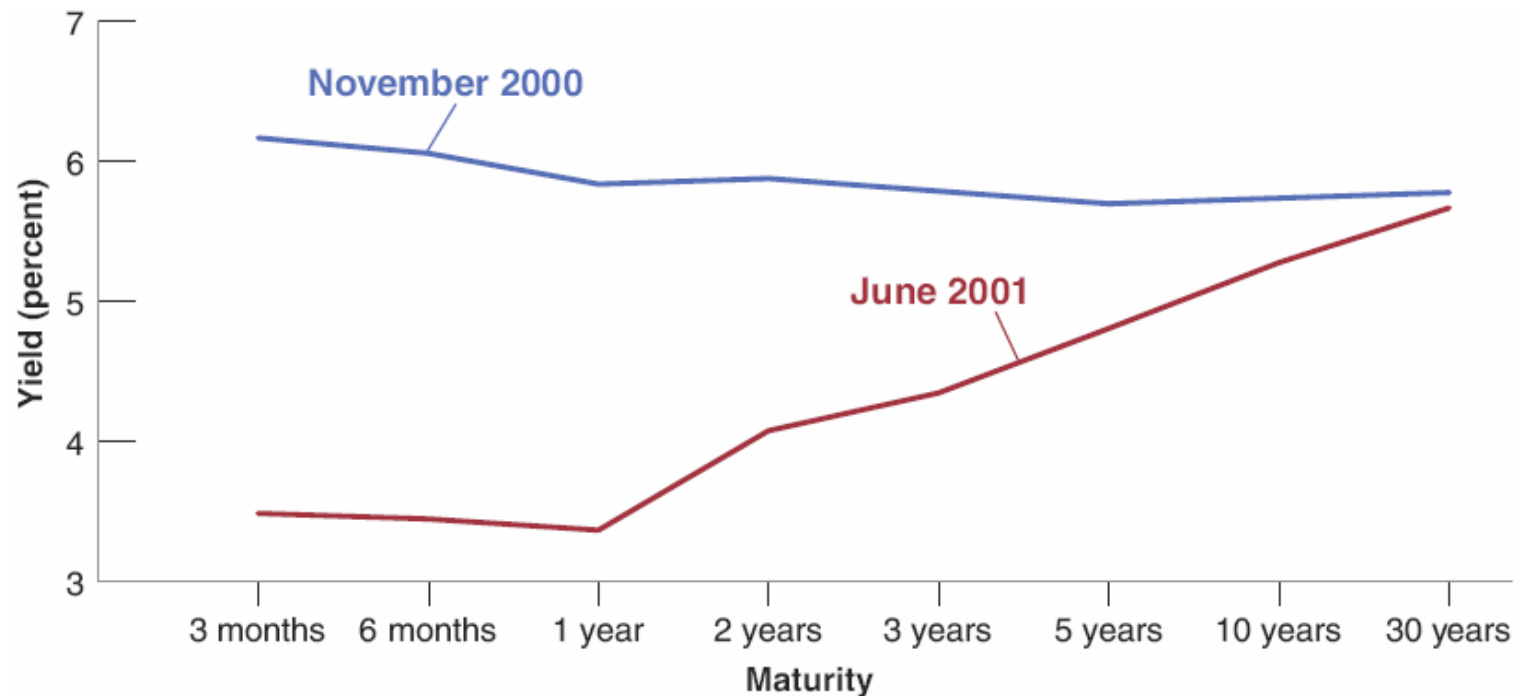
- **Yield to maturity or yield:** The interest rates associated with bonds of different maturities
- **Short-term interest rates:** Yields on bonds with a short maturity, typically a year or less
- **Long-term interest rates:** Yields on bonds with a longer maturity than a year
- **Term structure of interest rates or yield curve:** The relation between maturity and yield

14.2 Bond Prices and Bond Yields

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Figure 14.2 U.S. Yield Curves: November 1, 2000 and June 1, 2001

The yield curve, which was slightly downward sloping in November 2000, was sharply upward sloping seven months later.

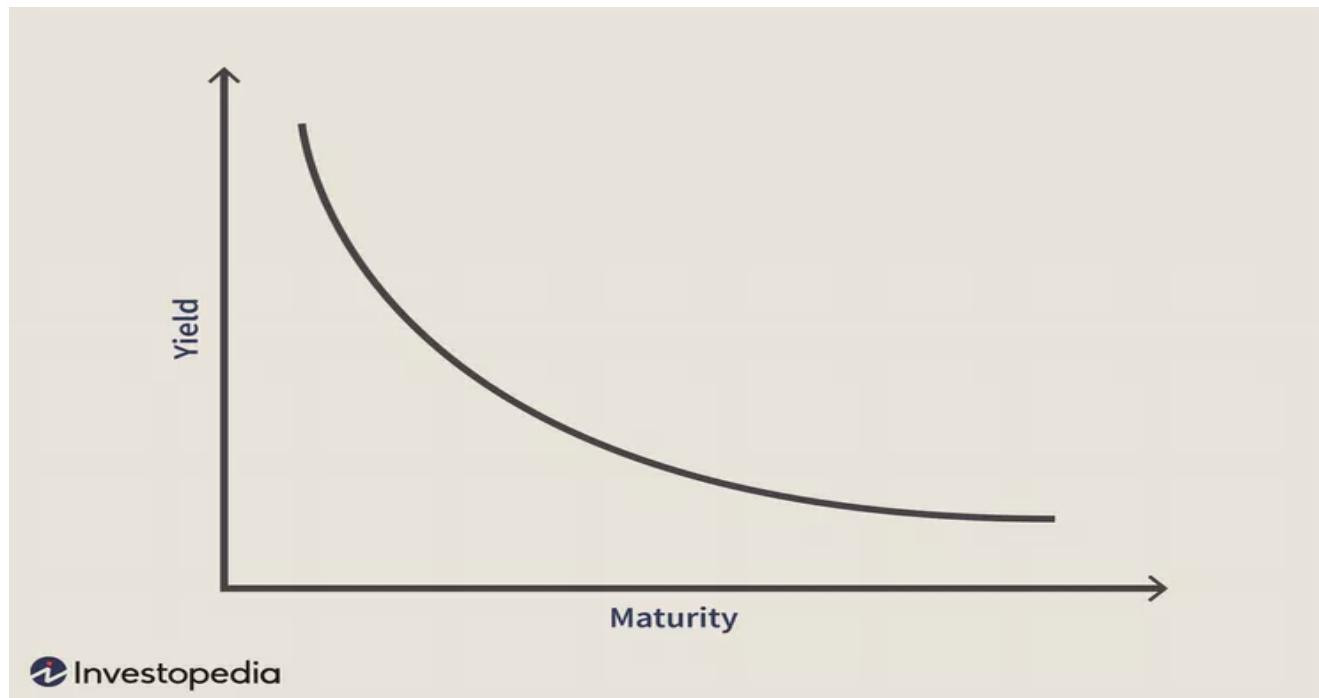


- *Source:* FRED. Series DGS1MO, DGS3MO, DGS6MO, DGS1, DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, DGS30.

Inverted Yield Curve

Inverted yield curves occur when short-term interest rates exceed long-term rates. Under normal circumstances, the yield curve is not inverted since debt with longer maturities typically carry higher interest rates than nearer-term ones.

The inverted curve, sometimes referred to as a negative yield curve, has proven in the past to be **a reliable indicator of a recession**



FOCUS: The Vocabulary of Bond Markets (1 of 3)

- **Government bonds:** Bonds issued by the governments
- **Corporate bonds:** Bonds issued by firms
- **Bond ratings:** ratings for default risk
- **Risk premium:** The difference between the interest rate paid on a given bond and the interest rate on the bond with the best rating
- **Junk bonds:** Bonds with high default risk
- **Discount bonds:** Bonds that promise a single payment at maturity called the **face value**

FOCUS: The Vocabulary of Bond Markets (2 of 3)

- **Coupon bonds:** Bonds that promise multiple payments before maturity and one payment at maturity
- **Coupon payments:** The payments before maturity
- **Coupon rate:** The ratio of the coupon payments to the face value
- **Current yield:** The ratio of the coupon payment to the price of the bond
- **Life:** The amount of time left until the bond matures
- **Treasury bills (T-bills):** U.S. government bonds with a maturity up to a year

FOCUS: The Vocabulary of Bond Markets (3 of 3)

- **Treasury notes:** U.S. government bonds with a maturity of 1 to 10 years
- **Treasury bonds:** U.S. government bonds with a maturity of 10 or more years
- **Term premium:** The premium associated with longer maturities
- **Indexed bonds:** Bonds that promise payments adjusted for inflation
- **Treasury Inflation Protected Securities (TIPS):** Indexed bonds introduced in the United States in 1997

14.2 Bond Prices and Bond Yields

(4 of 9)

- The price of a one-year bond that promises to pay \$100 next year: 一年期的現值

$$\$P_{1t} = \frac{\$100}{1 + i_{1t}} \quad (14.4)$$

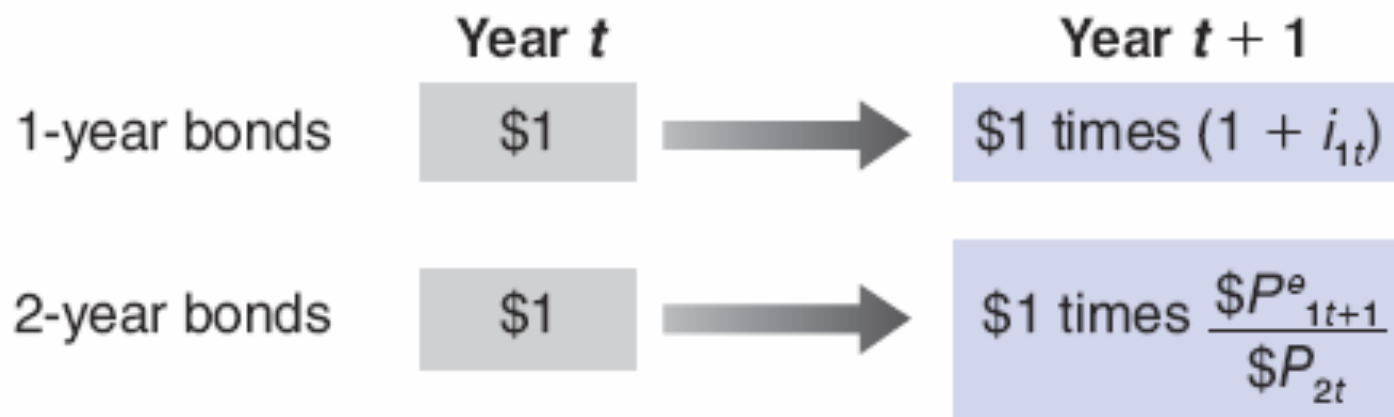
- The price of a two-year bond that promises to pay \$100 in two years: 二年期的現值

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)} \quad (14.5)$$

14.2 Bond Prices and Bond Yields

(5 of 9)

Figure 14.3 Returns from Holding One-Year and Two-Year Bonds for One Year



14.2 Bond Prices and Bond Yields

(6 of 9)

- **Arbitrage:** The expected returns on two assets must be equal.
- **Expectations hypothesis:** Investors care only about the expected returns and do not care about risk.
- Two bonds in Figure 14.3 must offer the same expected one-year return:

$$1 + i_{1t} = \frac{\$P_{1t+1}^e}{\$P_{2t}} \quad (14.6)$$

$$\$P_{2t} = \frac{\$P_{1t+1}^e}{1 + i_{1t}} \quad (14.7)$$

which means that the price of a two-year bond today is the present value of the expected price of the bond next year.

14.2 Bond Prices and Bond Yields

(7 of 9)

- The expected price of one-year bonds next year with a payment of \$100:

$$\$P_{1t+1}^e = \frac{\$100}{(1 + i_{1t+1}^e)} \quad (14.8)$$

so that

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)} \quad (14.9)$$

which is the same as equation (14.5)

14.2 Bond Prices and Bond Yields

(8 of 9)

- The *yield to maturity* on an n -year bond (**n -year interest rate**) is the constant annual interest rate that makes the bond price today equal to the present value of future payments on the bond.
- The yield to maturity on a two-year bond that satisfies:

$$P_{2t} = \frac{\$100}{(1 + i_{2t})^2} \quad (14.10)$$

is: $(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1t+1}^e)$

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e) \quad (14.11)$$

which means that *the two-year interest rate is (approximately) the average of the current one-year interest rate and next year's expected one-year interest rate.*

14.2 Bond Prices and Bond Yields

(9 of 9)

$$1 + i_{1t} = \frac{\$P_{1t+1}^e}{\$P_{2t}}$$

- Equation (14.8) with a risk premium x on the two-year bond:

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e + x)} \quad (14.12)$$

so that the two-year yield is:

$$i_{2t} \approx \frac{1}{2} (i_{1t} + i_{1t+1}^e + x) \quad (14.13)$$

which is the average of the current and expected one-year rate plus a risk premium.

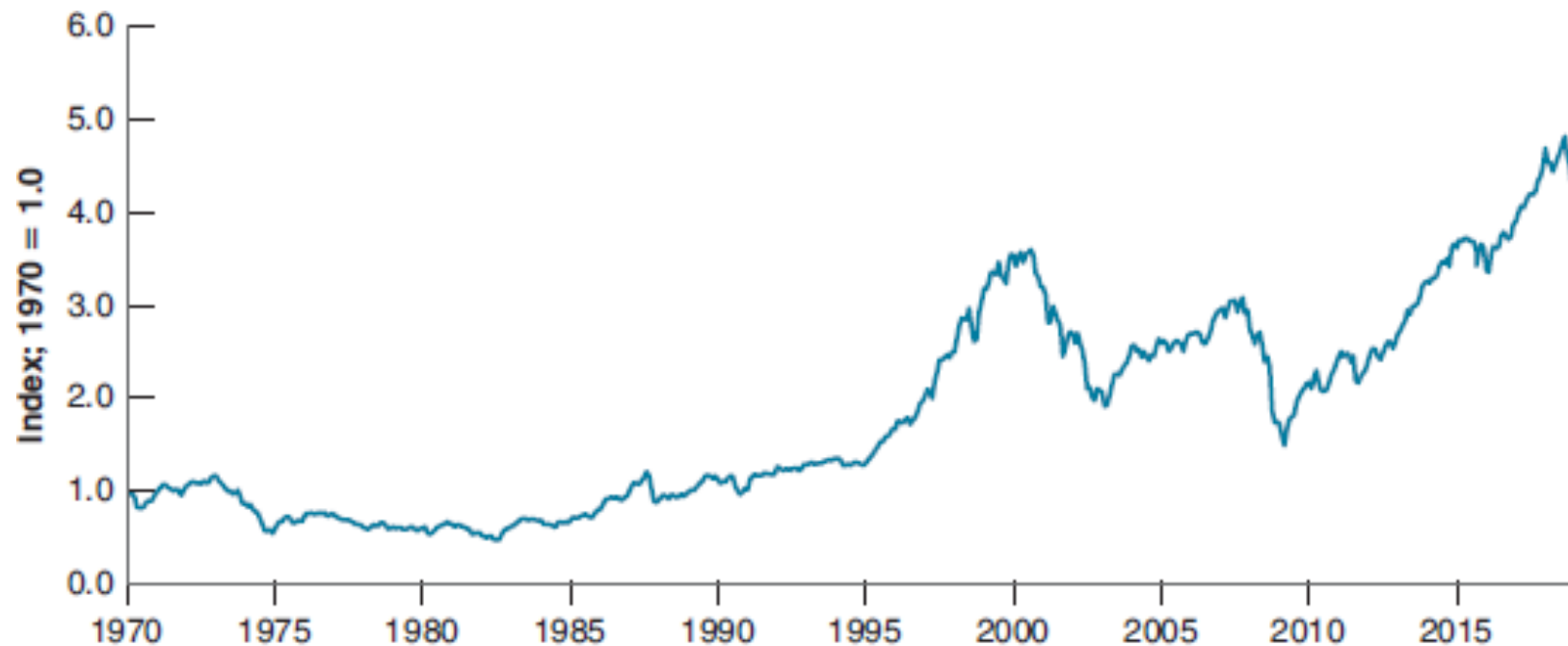
14.3 The Stock Market and Movements in Stock Prices (1 of 8)

- Firms finance themselves through:
 - **Internal finance:** Using their own earnings
 - **External finance:** Bank loans
 - **Debt finance:** Bonds and loans
 - **Equity finance:** Issuing **stocks** or **shares** that pay dividends

14.3 The Stock Market and Movements in Stock Prices (2 of 8)

Figure 14.4 Standard and Poor's Stock Price Index in Real Terms since 1970

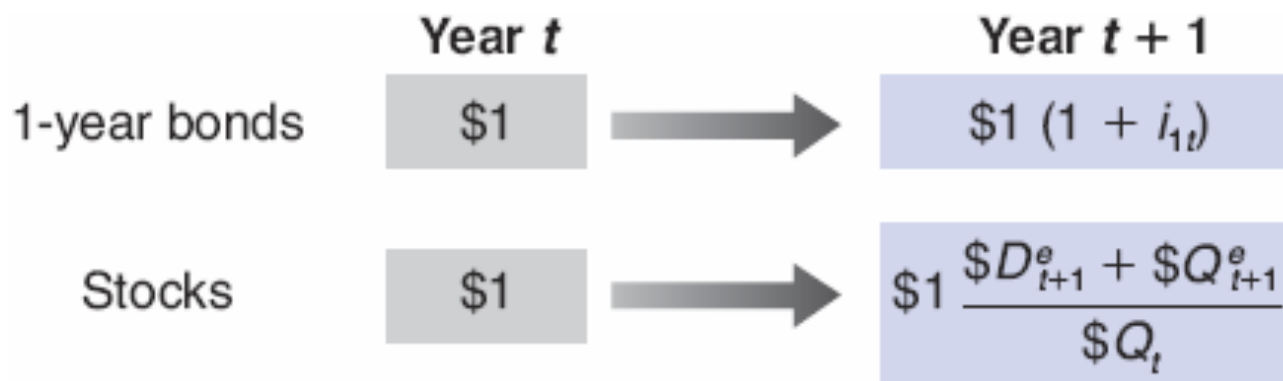
Note the sharp fluctuations in stock prices since the mid-1990s.



Source: FRED SP500, CPIAUCSL.

14.3 The Stock Market and Movements in Stock Prices (3 of 8)

Figure 14.5 Returns from Holding One-Year Bonds or Stocks for One Year



- $\$Q$ is the price of the stock
- $\$D^e$ is the expected dividend
- **Ex-dividend price:** The stock price after the dividend has been paid this year

14.3 The Stock Market and Movements in Stock Prices (4 of 8)

- Equilibrium requires that the expected rate of return from holding stocks for one year (left side) be the same as the rate of return on one-year bonds plus the equity premium x (right side):

$$\frac{\$D_{t+1}^e + \$Q_{t+1}^e}{\$Q_t} = 1 + i_{1t} + x$$

or

$$\$Q_t = \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$Q_{t+1}^e}{(1 + i_{1t} + x)} \quad (14.14)$$

$$\$Q_{t+1} = \frac{\$D_{t+2}^e}{(1 + i_{t+1} + x)} + \frac{\$Q_{t+2}^e}{(1 + i_{t+1} + x)}$$

14.3 The Stock Market and Movements in Stock Prices (5 of 8)

- If the expected prices in n years equal the present values of the expected prices and dividends:

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\ & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} + \frac{\$Q_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} \end{aligned} \quad (14.15)$$

- If the interest rate is positive, then it reduces to:

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\ & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} \end{aligned} \quad (14.16)$$

$\$Q_{t+n}^e$ approaches to zero as n goes to infinite

14.3 The Stock Market and Movements in Stock Prices (6 of 8)

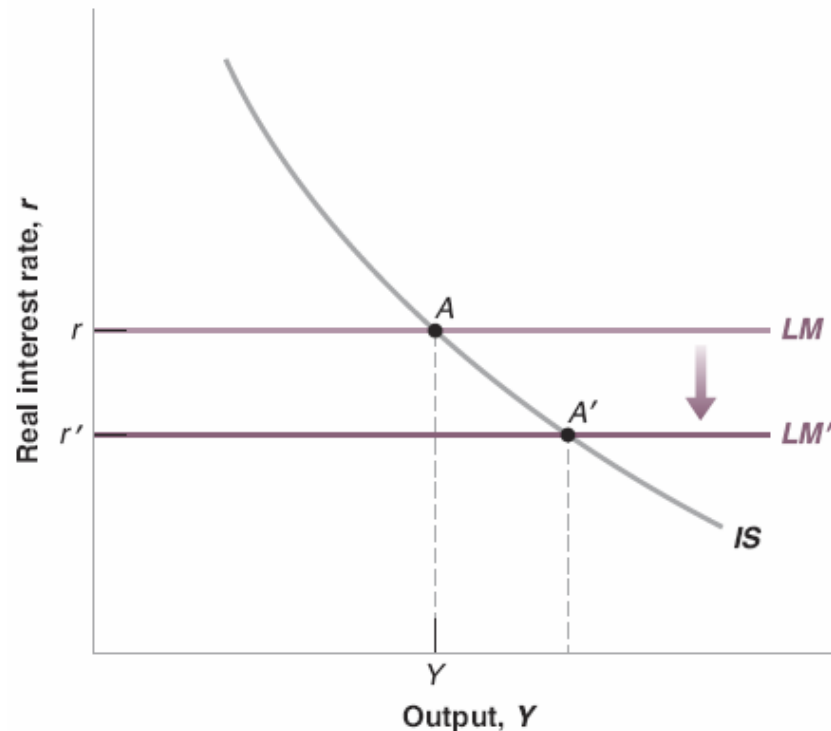
- Replacing the *nominal* interest rates with the *real* interest rates, then the real stock price is:

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots \quad (14.17)$$

- Implications:
 - Higher expected future real dividends lead to a higher real stock price.
 - Higher current and expected future one-year real interest rates lead to a lower real stock price.
 - A higher equity premium leads to a lower stock price.

14.3 The Stock Market and Movements in Stock Prices (7 of 8)

Figure 14.6 An Expansionary Monetary Policy and the Stock Market

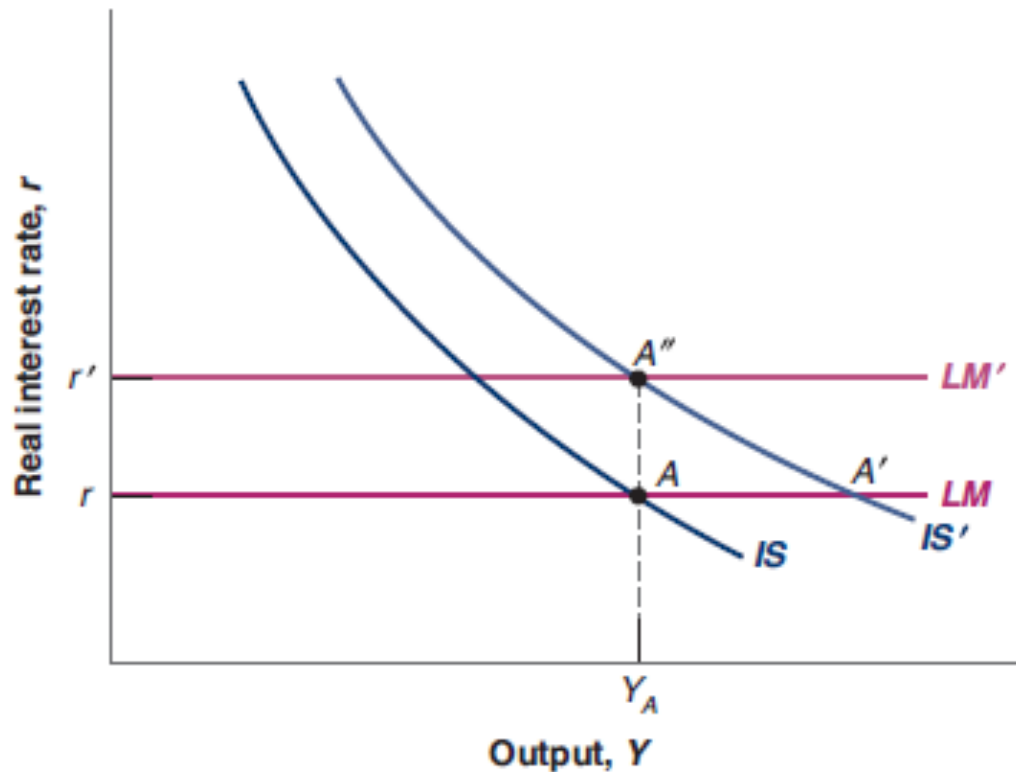


A monetary expansion decreases the interest rate and increases output. What it does to the stock market depends on whether or not financial markets anticipated the monetary expansion.

14.3 The Stock Market and Movements in Stock Prices (8 of 8)

Figure 14.7 An Increase in Consumption Spending and the Stock Market

An increase in consumption leads to higher level of output. What happens to the stock market depends on the reaction of the Fed.



FOCUS: Making (Some) Sense of (Apparent) Nonsense: Why the Stock Market Moved Yesterday, and Other Stories

- April 1997. Good news on the economy, leading to an increase in stock prices: “Bullish investors celebrated the release of market-friendly economic data by stampeding back into stock and bond markets, pushing the Dow Jones Industrial Average to its second-largest point gain ever and putting the blue-chip index within shooting distance of a record just weeks after it was reeling.”
- December 1999. Good news on the economy, leading to a decrease in stock prices: “Good economic news was bad news for stocks and worse news for bonds ... The announcement of stronger-than-expected November retail-sales numbers wasn’t welcome. Economic strength creates inflation fears and sharpens the risk that the Federal Reserve will raise interest rates again.”

14.4 Risk, Bubbles, Fads, and Asset Prices

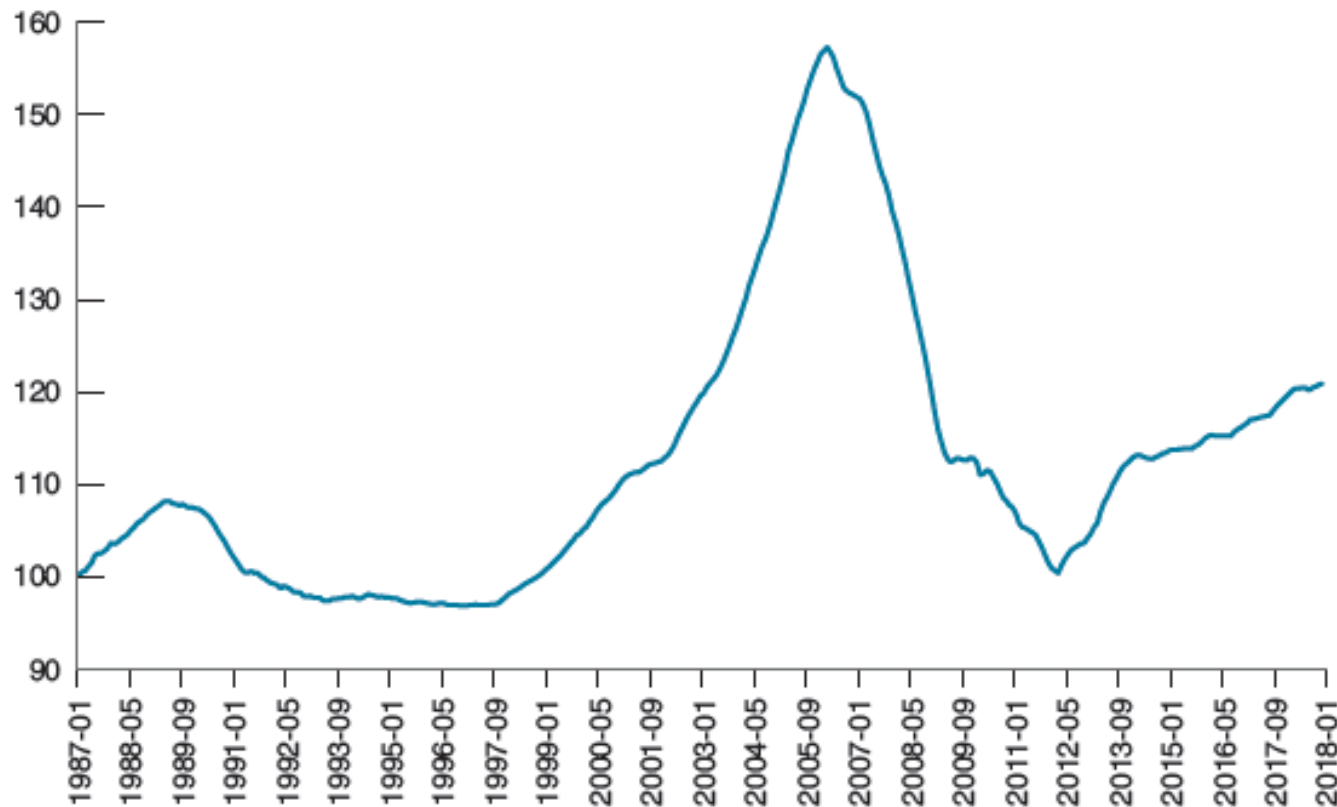
- **Fundamental value:** The present value of expected dividends given in equation (14.17) and that stocks are sometimes underpriced or overpriced.
- **Rational speculative bubbles:** Stock prices increase just because investors expect them to.
- **Fads:** Stocks become high priced for no reason other than its price has increased in the past.

FOCUS: Famous Bubbles: From Tulipmania in 17th-Century Holland to Russia in 1994

- In 1634, a “tulip bubble” began to take place as the price of rare bulbs started increasing, and speculators bought tulip bulbs in anticipation of even higher prices later.
- The price of a bulb called “Admiral Van de Eyck” jumped from 1,500 guineas in 1634 to 7,500 guineas in 1637.
- In 1994, a Russian “financier” created a company called MMM and promised shareholders a rate of return of at least 3,000% a year.
- Even though MMM was not involved in any type of production and held no assets, its share prices increased from 1,600 rubles in February to 105,000 in July.

FOCUS: The Increase in U.S. Housing Prices: Fundamentals or Bubble?

Figure 1 The U.S. Housing Price-to-Rent Ratio since 1987



Source: FRED: CSUSHPIISA: Case-Shiller Home Price Index. CUSR0000SEHA, Rent of Primary Residence.

APPENDIX: Deriving the Expected Present Discounted Value Using Real or Nominal Interest Rates

- Equations (14.1) and (14.3) are equivalent in their ways of expressing present discounted values.
- Divide both sides of equation (14.1) by the current price (P_t):

$$\frac{\$V_t}{P_t} = \frac{\$z_t}{P_t} + \frac{1}{1+i_t} \frac{\$z_{t+1}^e}{P_t} + \frac{1}{(1+i_t)(1+i_{t+1}^e)} \frac{\$z_{t+2}^e}{P_t} + \dots \quad (14.A1)$$

- Recall equation (14.3):

$$V_t = z_t + \frac{1}{1+r_t} z_{t+1}^e + \frac{1}{(1+r_t)(1+r_{t+1}^e)} z_{t+2}^e + \dots \quad (14.3)$$

- Each term on the right side of equation (14.3) is equal to the corresponding term in equation (14.A1).